ECE 285: Deep Generative Models — Homework 1

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Abstract

A convolutional VAE was trained on 5,216 chest X-rays without class labels. With 50 epochs, the validation performance is the best at epoch 45. The evaluation uses 624 samples and 624 held-out images. It gives FID of 283.72 and Inception Score of 2.095. The model is able to reproduces the overall chest shape from the input samples, but anatomical detail remains blurred. The code is at: https://github.com/shuoqihuang07/ECE285DeepGenerativeModels, in the HW1_VAE folder.

1 Input Space and Model Configurations

We use the Kaggle Chest X-Ray Images dataset to do the training, validation, and testing process for our VAE. In total, there are 5,216 images for training, 16 images for validation, and 624 images for testing. Both classes are pooled and resized to 128×128 and applied the grayscale in $[0, 1]$, without labels or augmentation.

Four 4×4 stride-2 convolutions with 32/64/128/256 channels and ReLU feed two linear heads for the mean and log variance of a 128-D diagonal Gaussian. A linear projection and four transposed convolutions decode its samples. Reparameterization gives $z = \mu + \exp(\frac{1}{2} \log \sigma^2) \odot \epsilon$, where $\epsilon \sim \mathcal{N}(0, I)$. Training minimizes the loss function, which includes the binary cross-entropy loss function over pixels and the KL-divergence loss over latent dimensions. The loss function of this VAE model is defined as:

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)}[\mathrm{BCE}(x, \sigma(f_\theta(z)))] + \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \| \mathcal{N}(0, I)). \tag{1}$$

Here's the summary of hyperparameter settings:

Optimizer	Adam	Learning rate	10^{-3}
Batch size	64	Epochs	50
Latent size	128	KL weight β	1
Training seed	42	Adam (β_1, β_2)	$(0.9, 0.999)$

2 Training

Loss falls quickly at first, then converges (Figure 1). Training loss drops from 10,411.04 to 9,431.28 nats per image. Among all the epochs, epoch 45 is selected due to the highest validation performance, with an error of 9,059.32.

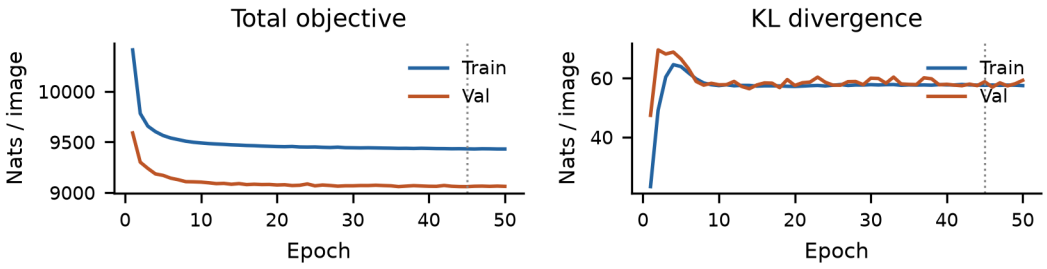


Figure 1: Losses over 50 epochs. The dotted line marks the selected model at epoch 45.

3 Output and Model Evaluation

The selected decoder generates 624 images from $z \sim \mathcal{N}(0, I)$. Figure 2 shows the first 16 draws of real images and generated images.

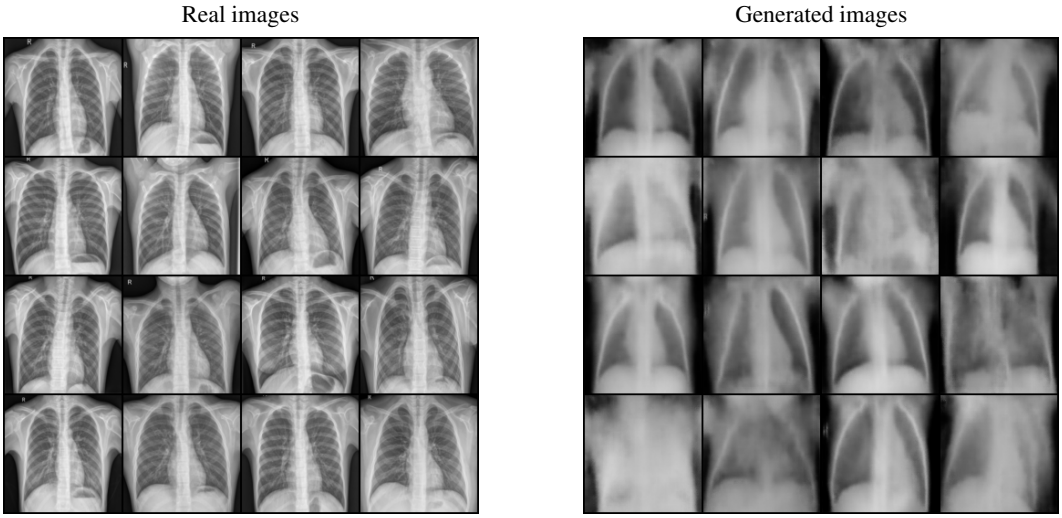


Figure 2: Real images (left) and generated images (right).

For the evaluation process, we use the inception score and FID score for generated images. Real and generated 8-bit PNGs receive the same preprocessing, which includes three identical channels, PIL bilinear resizing to 299×299, and ImageNet mean or standard normalization.

Here's a summary for the evaluation metrics:

Metric (624 real / 624 generated images)	Measured value
FID, pooled 2,048-D features ↓	283.72
Inception Score, generated, full set ↑	2.095
Inception Score, generated, 10 splits (mean ± SD)	2.045 ± 0.188
Inception Score, real test, full set (context only)	2.284

4 Limitations

The samples preserve the overall lung outlines, but with blurred ribs and boundaries. The lower validation loss might be insufficient since the validation set contains only 16 balanced images. Also, FID's covariance matrices are singular at $n = 624$. Therefore, for future improvements, increasing the validation set and running multiple training runs with different initial weights might help with reliability and robustness.